Extraction of Metals

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Extraction of Metals

Metals and ores

- The Earth's crust contains metals and metal compounds
- Useful metals are often chemically combined with other substances in rocks called ores
  - A metal ore is a rock that contains enough of the metal to make it worthwhile extracting

Why extraction is a reduction process

- Many metal ores are oxides
- Extracting the metal involves **removing oxygen** from the compound
- This means extraction is a **reduction** process
- Examples of oxide ores include:
  - Iron ore: hematite
  - Aluminium ore: bauxite

Unreactive metals may be found native

- Some very unreactive metals are found as uncombined elements
  - This is because they do not easily react with other substances
- These are called native metals
- Examples include gold and platinum

How the reactivity series links to extraction

- The position of a metal in the reactivity series affects how difficult it is to extract
- The higher a metal is in the reactivity series, the more difficult it is to extract from its ore
- Metals **above carbon** must be extracted using **electrolysis**
  - The most common example of a metal extracted by electrolysis is aluminium
- Metals **below carbon** can be extracted by **heating with carbon or carbon monoxide**

Stuck on something?

Ask questions and get help with exactly what you're revising.

Got it

- Metals lower in the reactivity series are easier to obtain than metals higher up the series

The extraction method depends on the position of a metal in the reactivity series

Metal	Method of Extraction
Most reactive	
Potassium	Extracted by electrolysis of molten compounds (chlorides or oxides) Large amount of electricity required so expensive process
Sodium	
Lithium	
Calcium	
Magnesium	
Aluminium	
Carbon	
Zinc	Extracted by heating with a reducing agent such as carbon or carbon monoxide
Iron	
Hydrogen	
Copper	Very unreactive metals Often found as the uncombined element or require little processing to extract
Silver	
Gold	
Least reactive	

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